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Subject: Programming with Java

ASSIGNMENT No: 4

TITLE: Wrapper Classes in Java

Problem Statement

Develop a Java application demonstrating the use of wrapper classes for conversion between primitive data types and objects, and performing basic operations on wrapped values.

Learning Objectives

To perform conversion between primitive data types and wrapper objects using Java wrapper classes.

Programme

```
J StudentMarks.java > ...
1  class StudentMarks {
    Run | Debug
2      public static void main(String[] args) {
3
4          String mark1 = "85";
5          String mark2 = "90";
6          String mark3 = "78";
7
8          // Convert String to Integer (Wrapper Class)
9          Integer m1 = Integer.valueOf(mark1);
10         Integer m2 = Integer.valueOf(mark2);
11         Integer m3 = Integer.valueOf(mark3);
12
13         int total = m1 + m2 + m3;
14
15         System.out.println("Marks:");
16         System.out.println("Subject 1: " + m1);
17         System.out.println("Subject 2: " + m2);
18         System.out.println("Subject 3: " + m3);
19         System.out.println("Total Marks: " + total);
20     }
21 }
```

```

J EmployeePayroll.java > EmployeePayroll
1 import java.util.Scanner;
2
3 class EmployeePayroll {
4     Run | Debug
5     public static void main(String[] args) {
6
7         Scanner sc = new Scanner(System.in);
8
9         System.out.print(s: "Enter Employee ID: ");
10        String empId = sc.nextLine();
11
12        System.out.print(s: "Enter Basic Salary: ");
13        String basicSalary = sc.nextLine();
14
15        System.out.print(s: "Enter Bonus Amount: ");
16        String bonus = sc.nextLine();
17
18        // Convert String to Wrapper Objects
19        Integer id = Integer.valueOf(empId);
20        Double salary = Double.valueOf(basicSalary);
21        Double bonusAmount = Double.valueOf(bonus);
22
23        // Validation
24        if (salary < 0 || bonusAmount < 0) {
25            System.out.println(x: "Invalid salary or bonus amount.");
26        } else {
27            double netSalary = salary + bonusAmount;
28
29            System.out.println(x: "\nEmployee Details");
30            System.out.println("Employee ID: " + id);
31            System.out.println("Basic Salary: " + salary);
32            System.out.println("Bonus: " + bonusAmount);
33            System.out.println("Net Salary: " + netSalary);
34        }
35
36        sc.close();
37    }

```

Output

Marks:

Subject 1: 85

Subject 2: 90

Subject 3: 78

Total Marks: 253

Enter Employee ID: 102922

Enter Basic Salary: 750000

Enter Bonus Amount: 12000

Employee Details

Employee ID: 102922

Basic Salary: 750000.0

Bonus: 12000.0

Net Salary: 762000.0

Conclusion

This assignment demonstrated the use for wrapper classes in java for converting primitive data types and strings into objects. It also showed how wrapper methods can perform arithmetic operations and validate user input efficiently. Overall, the program improved the understanding of type conversion, object handling and the practical applications of wrapper classes in java programming.